VISVESVARAYA TECHNOLOGICAL UNIVERSITY

"JNANA SANGAMA", BELAGAVI - 590 018

A PROJECT REPORT

on

"Credit Card Fraud Detection using Machine Learning Model"

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In partial fulfillment of the requirements for the VIII semester

BACHELOR OF ENGINEERING

in

INFORMATION SCIENCE & ENGINEERING

Under the Guidance of

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at

SAHYADRI

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2018 - 19

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CERTIFICATE

This is to certify that the work of project entitled **"Credit Card Fraud Detection using Machine Learning Model"** has been carried out by **Meghna N (4SF15IS047)**, the bonafide student of Sahyadri College of Engineering & Management in partial fulfillment of the requirements for the VIII semester of Bachelor of Engineering in Information Science & Engineering of Visvesvaraya Technological University, Belagavi during the year 2018 - 19. It is certified that all corrections/suggestions indicated for Internal Assessment have been incorporated in the Report deposited in the departmental library. The project report has been approved as it satisfies the academic requirements in respect of project work prescribed for the said degree.

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DECLARATION

I hereby declare that the entire work embodied in this Project Report titled **"Credit Card Fraud Detection using Machine Learning Model"** has been carried out by me at Sahyadri College of Engineering and Management, Mangaluru under the supervision of **Ms. Asha B. Shetty,** for the award of **Bachelor of Engineering** in **Information Science & Engineering**. This report has not been submitted to this or any other University for the award of any other degree.

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Abstract

The project is on credit card fraud detection using machine learning models. As fraudster are increasing day by day and transaction are done by credit card and there is various types of fraud. So to solve this problem the study of the machine learning models is done and to determines the model which effectively detects credit card fraud and prompt the user or combination of the model. By this transaction is tested individually and whatever suits the best is further proceeded and foremost goal is to detect fraud by filtering the above models to get better result.

Acknowledgement

It is with great satisfaction that I am submitting the Project Report on **"Credit Card Fraud Detection using Machine Learning Model"**. I have completed it as a part of the curriculum of Visvesvaraya Technological University, Belagavi in partial fulfillment of the requirements for the VIII semester of Bachelor of Engineering in Information Science & Engineering.

I am profoundly indebted to our guide, **Ms. Asha B. Shetty**, Assistant Professor, Department of Information Science & Engineering for innumerable acts of timely advice, encouragement and I sincerely express my gratitude.

I also thank **Mr. Rithesh Pakkala P.**, Project Coordinator & Assistant Professor, Department of Information Science & Engineering for their constant encouragement and support extended throughout.

I express my sincere gratitude to **Dr. Shamanth Rai**, Head & Associate Professor, Department of Information Science & Engineering for his invaluable support and guidance.

I sincerely thank **Dr. R. Srinivasa Rao Kunte**, Principal, Sahyadri College of Engineering & Management and **Dr. D. L. Prabhakara**, Director, Sahyadri Educational Institutions, who have always been a great source of inspiration.

Finally, yet importantly, I express my heartfelt thanks to my family & friends for their wishes and encouragement throughout the work.

Meghna N (4SF15IS047)

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Chapter 1

Introduction

Machine Learning is a field of concentrate that enables PCs to learn without being expressly modified. Rather than composing code, you feed information to the generic algorithm, and it fabricates rationale dependent on the information given. This venture forestalls Credit card extortion by identifying the strange conduct of exchanges. Here we store the ordinary action of the clients exchange, if there is an abnormal exchanges, it will identify.

Credit card misrepresentation recognition is a pertinent issue that draws the consideration of AI and computational insight networks, where countless arrangements. Indeed,this issue gives off an impression of being especially testing from a learning viewpoint,since it is described in the meantime by class imbalance, namely, genuine exchanges fardwarf cheats, and idea float, to be specific, exchanges may change their factual proper-ties over time. In a genuine world FDS, the enormous stream of installment demands isimmediately filtered via programmed devices that figure out which exchanges to approve. Classiers are regularly used to dissect all the approved exchanges and alarm the most suspicious ones.

Credit card extortion identification likewise has two other profoundly impossible to miss qualities. At first clearly the restricted time range in which the acknowledgment or dismissal choice must be made and then colossal measure of charge card tasks that must be handled at a given time. The circumstance in India is genuinely simple: more than 1.2 million Credit card activities happen in a given day, 98 level of them being taken care of on line. Obviously, simply not very many will be deceitful, however this fair implies the bundle where these needles are to be found is essentially gigantic.

These exchanges stay unlabeled until clients find and report fakes, or until a sufficient measure of time has passed with the end goal that non questioned exchanges are considered genuine. Fraud avoidance is critical to card guarantors, yet not making unrealistic or essentially awkward the day by day card use by a huge number of clients. At the point when all the time required for the remote associations and the fundamental activity handling is considered, a misrepresentation location framework more often than not has close to a somewhat little portion of one moment to play out its assignment. This allotted time probably won't be sufficient for huge database inquiries. Obviously, this might be lightened if uniquely configured and devoted equipment is accessible, however it will absolutely result in higher start up and support costs.

Chapter 2

Literature Survey

Literature survey is a base for reasearch through which learning can be picked up and furthermore depicts the different works done in the important field.

In [1], Andrea Dal Pozzolo *et al.* recognizing fakes in charge card exchanges is maybe a standout amongst the best test information for computational insight calculations. Truth be told, this issue includes various pertinent difficulties, in particular: idea float (clients' propensities develop and fraudsters change their techniques after some time), class irregularity (authentic exchanges far dwarf cheats), and verication inactivity (just a little arrangement of exchanges are auspicious checked by agents). Nonetheless, by far most of learning calculations that have been proposed for misrepresentation identification depend on suspicions that barely hold in a true extortion discovery framework (FDS). In this paper there is Study of versatile and conceivably nonlinear total techniques for the classifiers prepared on input's and postponed directed examples.

In [2], Cesare Alippi *et al.* presents a various leveled change-discovery tests (HCDTs), as viable online calculations for identifying changes in information streams. HCDTs are described by a various leveled design made out of a location layer and an approval layer. The discovery layer consistently investigates the info information streams by methods for an on the web, consecutive CDT, which works as a low-intricacy trigger that immediately identifies potential changes in the process creating the information. The approval layer is enacted when the identification one uncovers a change, and plays out an ofine, increasingly refined investigation on as of late gained information to diminish false alerts. HCDTs give a conceptual preparing level to make savvy frameworks versatile in powerful and advancing conditions.

In [3], Krishna Modi and Reshma Dayma Cashless exchanges, for example, online exchanges, Mastercard exchanges, and versatile wallet are winding up increasingly well known in budgetary exchanges these days. With expanded number of such cashless exchange, number of deceitful exchanges are likewise expanding. Extortion can be recognized by breaking down spending conduct of (clients) from past exchange information. Strategies to distinguish deceitful exchanges and correlation of these techniques. One of these or blend of these strategies can be utilized to distinguish false exchanges.

In [4], Shuo Wang *et al.* Online class awkwardness learning is another learning issue that consolidates the difficulties of both internet learning and class unevenness learning. It manages information streams having extremely skewed class disseminations. This sort of issues generally exists in true applications, for example, deficiency analysis of ongoing control checking frameworks and interruption discovery in PC systems. In our prior work, we dened class awkwardness on the web, and proposed two learning calculations OOB and UOB that construct a group model conquering class irregularity continuously through re testing and time-rotted measurements. This paper improved and examined top to bottom two gathering learning techniques OOB and UOB utilizing re inspecting and the time-rotted measurement to conquer class lopsidedness on the web.

In [5], Ayushi Agrawal *et al.* method for 'Charge card Fraud Detection' is created. As fraudsters are expanding step by step. What's more, fraudulent exchanges are finished by the Mastercard and there are different sorts of misrepresentation. So to take care of this issue blend of method is utilized like Genetic Algorithm, Behavior Based Technique and Hidden Markov Model. The principle worry of this examination was to identify least and exact false extortion discovery.

In [6], Masoumeh Zareapoor, Pourya Shamsolmoali contrasting a few standard classifiers and packing group classifier which is a novel system in charge card extortion discovery strategy. There are utilizing the Naive Bayes Classifiers to concoct the answer for the Mastercard misrepresentation.

In [7], Cesare Alippi *et al.* classiers work in developing situations by grouping occasions and responding to idea float. In stationary conditions, a JIT classier improves its precision after some time by abusing extra regulated data originating from the eld. In non-stationary conditions, notwithstanding, the classier responds when idea float is recognized; the present classication setup is disposed of and an appropriate one enacted to keep the precision high. The proposed JIT approach is a exible tooNaive Bayes Classifiers to think of the answer for the charge card fraud.l for planning versatile classiers ready to adapt to classication issues influenced by idea float.

In [8], Gregory Ditzler and Robi Polikar Learning in non-stationary conditions, otherwise called learning idea float, is worried about gaining from information whose measurable qualities change after some time. Idea float is additionally convoluted if the informational index is class imbalanced. While these two issues have been autonomously tended to, their joint treatment has been for the most part under investigated. We depict two gathering based methodologies for taking in idea float from imbalanced information. In this paper, we talk about taking in idea float from imbalanced information.

In [9], Mohammed Ibrahim Alowais and Lay-Ki Banking industry endures lost in a huge number of dollars every year brought about with Visa misrepresentation. Enormous exertion, time and cash have been spent to distinguish misrepresentation where there are considers done on making customized model for each Visa holder to recognize extortion. These investigations asserted that each card holder conveys distinctive spending conduct which requires customized model. Demonstrates the procedures associated with making customized models just as the Real Transaction Aggregated Model (RTAM). To make the customized models, we have utilized the exchanges of the three people separately.

In [10], Ryan Elwell,and Robi Polikar presenting a group of classiers-based methodology for gradual learning of idea float, described by non-stationary situations (NSEs), where the hidden information disseminations change after some time. The proposed calculation, named Learn++.NSE, gains from back to back groups of information without making any suppositions on the nature or rate of float; it can gain from such situations that experience steady or variable rate of float, expansion or cancellation of idea classes, just as patterned float. Gives the best execution: in online student correlation.

In [11], Abhinav Srivastava *et al.* Because of a fast headway in the electronic business innovation, the utilization of Visas has drastically expanded. As Mastercard transforms into the most noticeable technique for portion for both online similarly as typical purchase occasions of deception related with it are in like manner rising. Here, the model succession of tasks in charge card trade dealing with using HMM and show how it tends to be utilized for the discovery of cheats. Near investigations uncover that the Accuracy of the framework is near 80 percent over a wide variety in the info information. The framework is additionally versatilefor dealing with huge volumes of exchanges.

In [12], Janez Demsar(2006) While examinations utilizing a solitary informational collection are bothered by the one-sided fluctuation estimations because of conditions between the examples of precedents drawn from the informational index, in correlations over different informational index the change originates from the contrasts between the informational collections, which are typically free. There are utilizing the Wilcoxon marked position test and Friedman test for the measurable correlations of classifiers over numerous informational indexes.

In[13], Dimitris K. Tasoulis *et al.* While techniques for seeing two learning estimations on a lone informational collection have been examined for a long while as of now, the issue of measurable tests for examinations of more calculations on various informational collections, which is considerably increasingly fundamental to ordinary AI thinks about, has been everything except disregarded. This article audits the present practice and afterward hypothetically and experimentally looks at a few appropriate tests. Part thickness estimation is a built up non-parametric system that has been effectively utilized by the bunching network.

In [14], Emin Aleskerov *et al.* CARDWATCH, a database digging framework utilized for Visa misrepresentation recognition, is introduced. The framework depends on a neural system Learning module, gives an interface to an assortment of business databases and has an agreeable graphical UI. Test outcomes got for artificially produced charge card information and an auto-cooperative neural system model show exceptionally fruitful misrepresentation recognition rates. The framework is effectively extensible and ready to work straightforwardly on an enormous assortment of business databases.

In [15], Jose R. Dorronsoro *et al.* on-line framework for extortion discovery of Visa activities dependent on a neural classier. Since it is introduced in a value-based center for task conveyance, and not on a card-issuing organization, it acts exclusively on the data of the activity to be appraised and of its quick past history, and not on noteworthy databases of past cardholder exercises. Among the principle qualities of Master card trafc are the incredible awkwardness among legitimate and fake tasks, and an extraordinary level of blending between both. To guarantee legitimate model development, a nonlinear rendition of Fisher's discriminant investigation, which sufficiently isolates a decent extent of deceitful tasks from other closer to ordinary trafc, has been utilized. Frameworks, for example, Minerva demonstrate that neural, task based, continuous extortion recognition frameworks are in fact plausible, however exceptionally intriguing from an absolutely prudent perspective.

Chapter 3

Problem Definition

The Credit Card Fraud Detection Problem incorporates displaying past charge card exchange with the information of the ones that ended up being extortion. To execute diverse sort of AI model.

Chapter 4

Objectives

Following are the goals of the proposed task:

1. The project centers around examining diverse models and locate the most fitting strategy which help in limiting the extortion as well as help in future upgrade

2. It additionally centers around dissecting whether an exchange made is false or not, by utilizing AI systems

3. The framework is prepared by giving information tests and relying upon this information the result is produced

4. Utilizing the acquired yield we can arrive at the resolution whether exchange is false or not and in this way counteracting the extortion

5. The model will actualize quick and exact algorithm on gathered charge card exchange record from all the conceivable sources

Chapter 5

Software Requirements Specification

5.1 Introduction

The SRS is a report, which portrays totally the outside conduct of the delicate product. The as a matter of first importance work of a product engineer is to ponder the framework to be developed and indicate the client prerequisites before going for the arranging stage. This report will let us know how this framework carries on and reacts.

5.1.1 Purpose

The principle reason for this SRS is to decipher the thoughts in the psyches of a customer into a formal record. Through SRS the customer obviously depicts what it anticipates from the proposed framework and the engineer plainly comprehends what abilities are required to assemble the framework. The reason for this archive is to fill in as a manual for the designers and analyzers who are in charge of the improvement of the framework. Normally the diabetic patient comes to think about his infection at a later stage and it might have additionally caused numerous other related inconveniences. Thus anticipating the diabetes at a beginning period relying upon various parameters might be valuable.

5.1.2 Scope

This SRS portrays the need of the framework. It is intended for use by the clients and will be the reason for approving whether the exchange is deceitful or not.

5.1.3 Overview

This SRS report covers both majority and minority of software/hardware requirements, parameters of the project and goals. It contains full depiction of proposed purposes framework conduct and system requirements.

5.2 The Overall Description

This segment will give the data about the entire framework. The framework will be disclosed in its setting to indicate how precisely the framework connects with different frameworks and acquaint in its setting with show how precisely the framework interfaces with different frameworks and present its essential usefulness. It will likewise portray kind of partner that will utilize the framework and what usefulness that is accessible for each sort.

5.2.1 Product Perspective

The framework ought to be prepared with dataset. The framework takes distinctive components and numerous different credits to foresee the result.

5.2.2 Operating Environment

The framework is created in windows working framework utilizing python programming language.

5.2.3 Product Functions

The User interface is assembled where the client inputs his data and this data is put away in the framework. The client can likewise bring their data from the framework which is as of now inputted and after that can enter their certifications.

5.2.4 User Characteristics

Client should have fundamental PC learning and client spending designs just as normal client geographic areas to check his character.

5.2.5 Constraints

The accuracy of the framework will rely upon the kind of preparing input gave, the more precise the dataset the better the continuous examination will be. The nature of the outcome will likewise influence the examination.

5.2.6 Assumptions and Dependencies

Suspicion made is that the client necessities will be met and that the modules of the project will have network to more frauds that can be maintained a strategic distance from.

5.3 Specific Requirements

This area contains the majority of the practical and quality prerequisites of the framework. It gives depiction of the of the framework and every one of its highlights.

5.3.1 Hardware Requirements

- **Processor:** Intel(R) core(TM) i5
- **RAM:** 4.00 GB or more
- **Hard Disk:** 1 GB or more

5.3.2 Software Requirements

- **OS:** Windows 8.1 or more
- **Programming language:** Python
- **Platform:** Anaconda

5.4 Functional Requirements

The principle usefulness given by this undertaking is identify the false exchange and keep away from it.

- **Input :** Credit Card dataset to the application

- **Processing:** The application undergoes different algorithms
- **Output:** Detecting the fraudulent transaction and to prevent the transaction

5.5 Non Functional Requirements

1. **Performance:** The performance is measured by how accurately the application will detect the information
2. **Reliability:** The application must perform consistent prediction
3. **Availability:** The application must most likely identify the outcome at any instance of time
4. **Maintainability:** The application is simple, easy to use and user friendly

5.6 Design Constraints

- Suitable qualities ought to be considered
- The application shall be trained prior to its use
- The program shall be implemented using Jupyter Notebook on Anaconda platform
- The application shall display error messages to the user when an error is detected
- Finally if there is appropriate values and meets the different constraints of the algorithms the required result is displayed

Chapter 6

System Design

6.1 Architecture Diagram

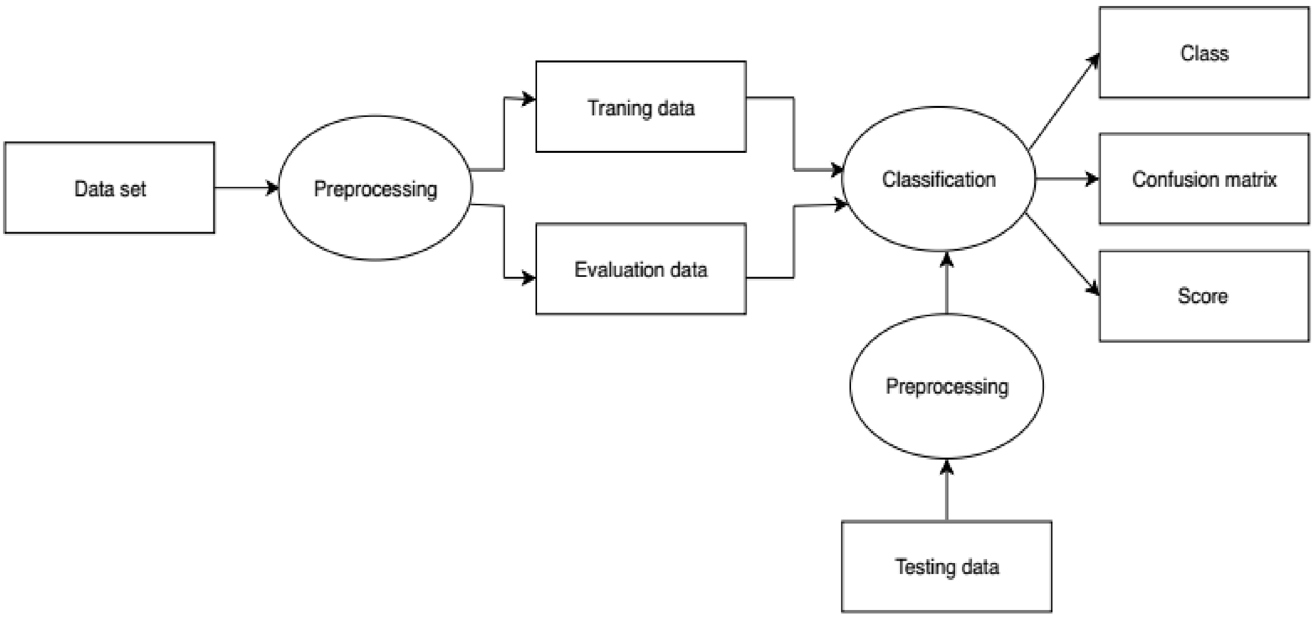

Figure 6.1: Architecture Diagram of Fraud Detection System

An engineering chart is a graphical portrayal that delineates a specific tale about a framework being described.The general design outline of Fraud Detection System utilizing Machine Learning Models is appeared in the figure. The graph incorporates stages like Pre-handling, preparing information, testing, assessing and checking whether the given informational collection is misrepresentation or not.

The outline portrays the general working of fraud detection system. At first the information is prepared or pre-handled subsequent to preparing we will get standardized information on which we can perform testing and assessing. In testing it will characterize the information and in assessment it will group and look at the information. Toward the end we will acquire perplexity network review store and to which class does the information have a place with accordingly.

6.2 Class Diagram

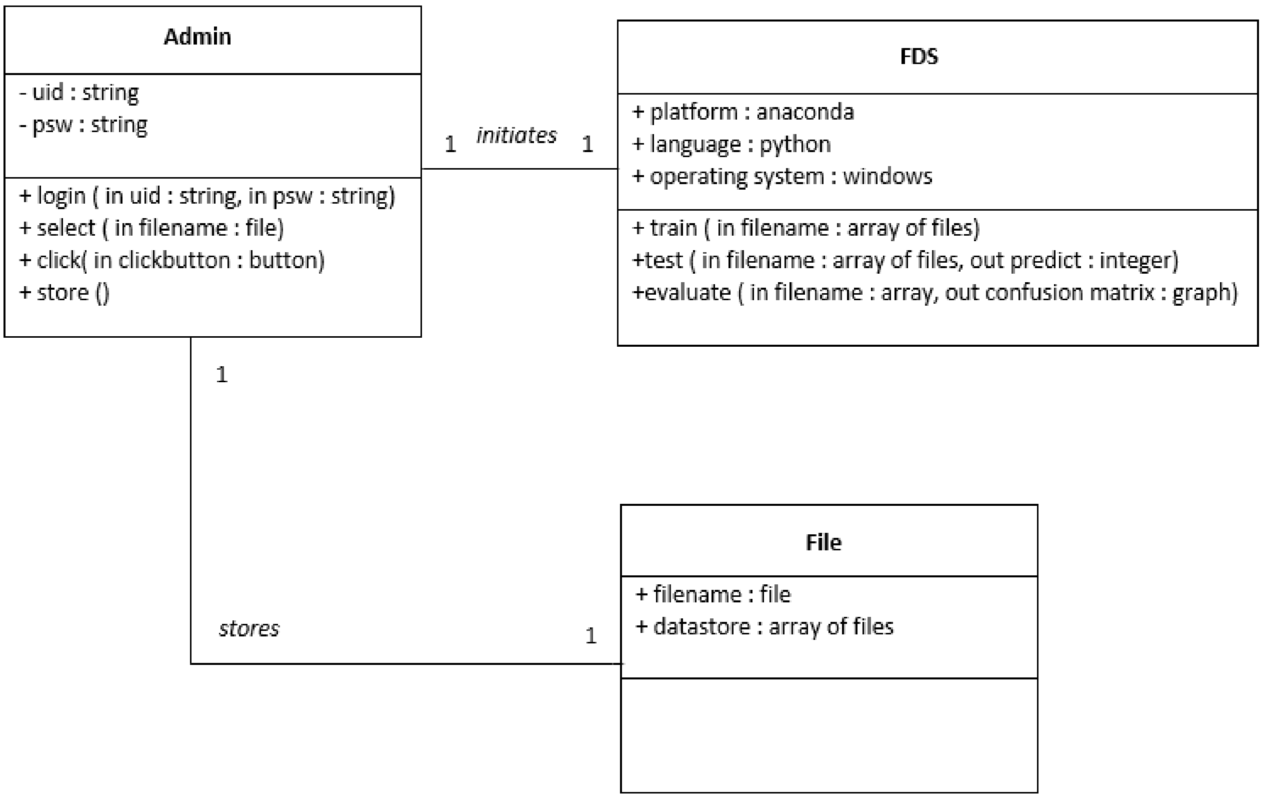

Figure 6.2: Class Diagram of Fraud Detection System

It is a general reasonable model that depicts the structure of a framework by demonstrating the framework's classes, their traits, alongside activities and the relationship among items. The general Class graph of the framework is appeared in the figure. It incorporates classes like Admin, FDS and File. Administrator is a parent class who starts every one of the activities on the framework. The class Admin has got quality uid which takes the id of the administrator and psw which takes the secret key .It has got activities like login, select, snap and store whose perceivability is public.

Class FDS has characteristics like Platform, language and working framework. It can perform different tasks like train, test and assess whose perceivability is open.
Here we accept File as a different class whose qualities are record name and information store which stores different documents and informational indexes.

6.3 State Diagram

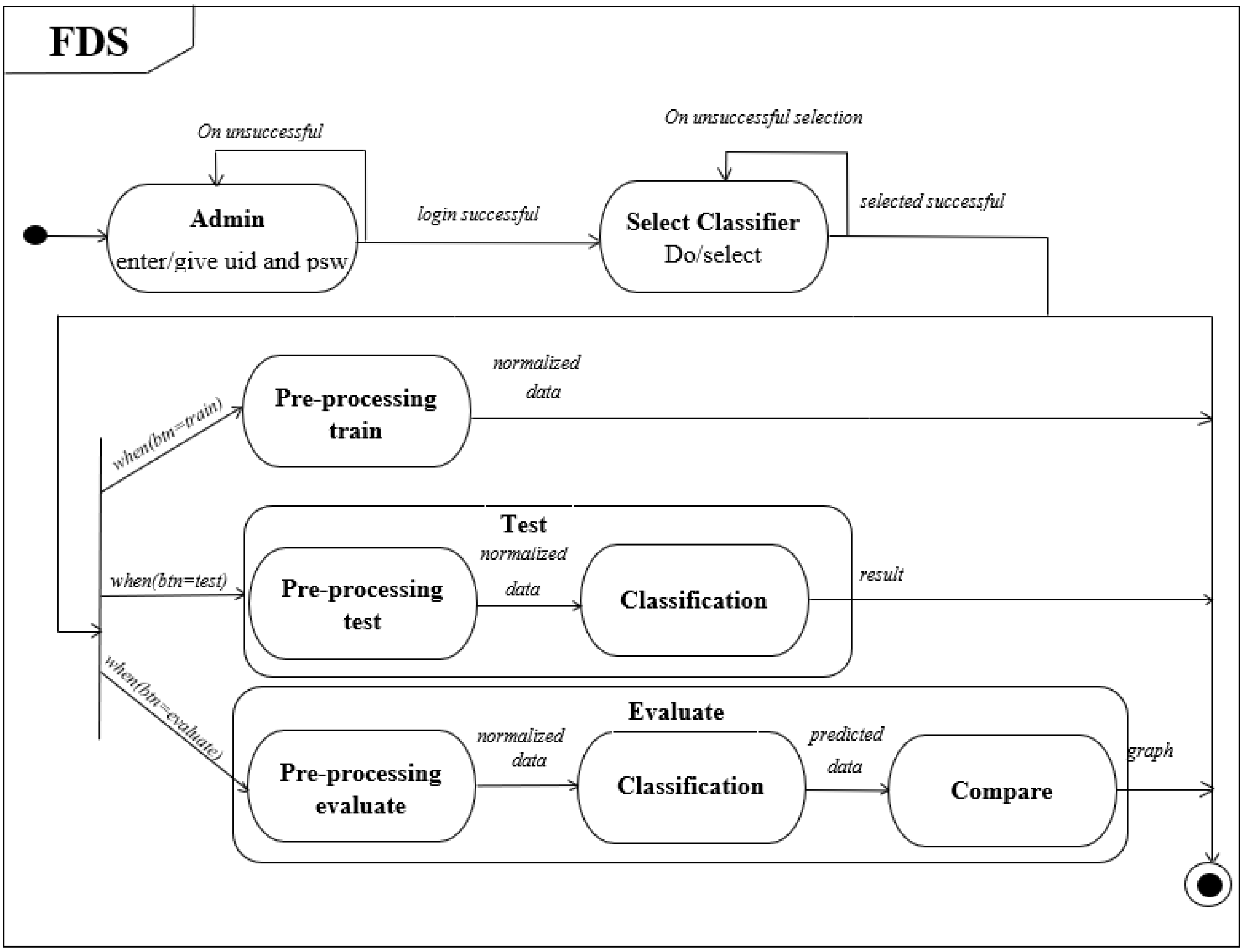

Figure 6.3: State Diagram of Fraud Detection System

A State chart demonstrates the progression of control starting with one state then onto the next. They delineate the conduct of classes in agreement to outside improvements. The State graph as appeared in the Figure 3.8 has got three stages i.e., Training ,Testing and assessing
At first the administrator needs to enter uid and secret key and on fruitful sign in it will enter the following stage where he needs to choose classifier, and on ineffective sign in it will remain in a similar state until right uid and secret phrase is given. Same if there should arise an occurrence of choosing classifier, on fruitless choice of classifier it

stays in a similar state. Next it goes to the pre-preparing stage where it will prepare the information.

On the chance that the client taps on testing catch it will go to pre-preparing stage and than arrangement organize where it will characterize the information. It will give the outcome to the client.

In the event that the client taps on assessing catch it will go to three unique stages like pre-handling ,arrangement and comparison.At the end it will show the diagram to he admin.At any stage the client can stop the tasks which drives the framework into stop organize.

6.4 Interaction Model

6.4.1 Use Case Model

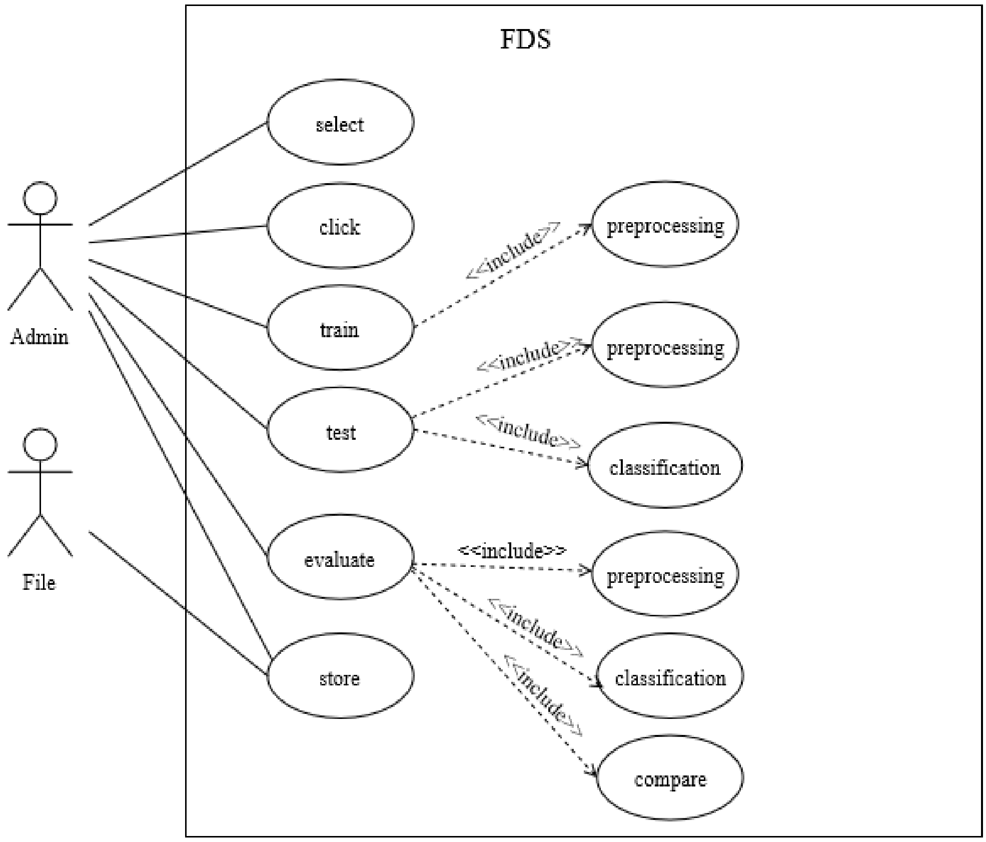

Figure 6.4: Use Case Diagram of Fraud Detection System

Figure 6.8 represents the use case diagram of Credit Card Fraud Detection System using Machine Learning Techniques. Here, the square shape box speaks to the framework and oval speak to the utilization cases. Stick man is only the outside clients and the lines show the interaction.Basically here there are two performing artists Admin and File. Administrator can choose, click, train, test, assess and store the data.The records are put away in documents. Here train incorporates pre-preparing, test incorporates pre-handling and grouping. Assess incorporates pre-handling and arrangement and correlation.

6.4.2 Sequence Diagram

Sequence diagrams illustrates the order in which the objects interact with each other.

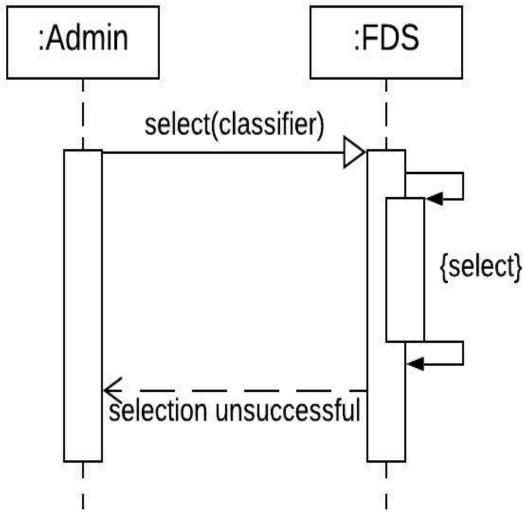

Figure 6.5: Sequence Diagram of Selecting Classifier

The strong bolts speak to the capacity call, dashed bolt speaks to affirmation and the last bolt speaks to messages.The protests here are Admin, Fraud Detection System (FDS) and File.

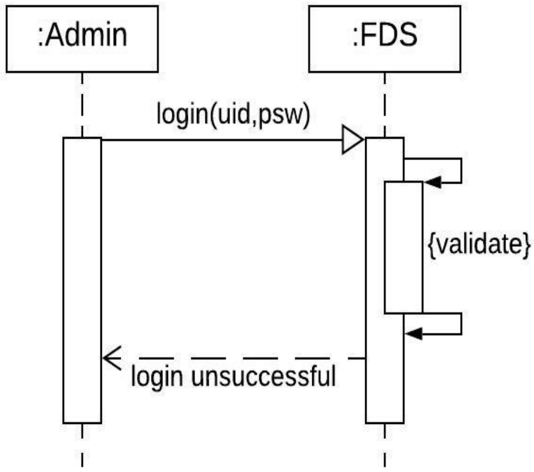

Figure 6.6: Sequence Diagram of user login

The figure speaks to the arrangement of unsuccessful sign in, when the administrator enters the wrong administrator id (UId) and secret word the framework approves the secret phrase and returns the message as login ineffective.

The given figure speaks to the grouping chart for misrepresentation discovery framework. Here the administrator enters the uid and secret key and the framework approves the secret word and on right approval it gives the message sign in fruitful. After the fruitful sign in the administrator chooses the classifier by mentioning for classifier document. On the off chance that the choice is ineffective it will send an affirmation to client. It begins the preparation by offering direction to the framework.

The framework solicitations to recover the information from the document. The record restores the information to the framework. The information is pre-prepared. The wake of preparing the framework solicitations to recover the information from the document for testing. The wake of testing it will show the outcome to the client. For assessment it will demand the informational indexes from the record and it will arrange and contrast the information and return diagram with the client.

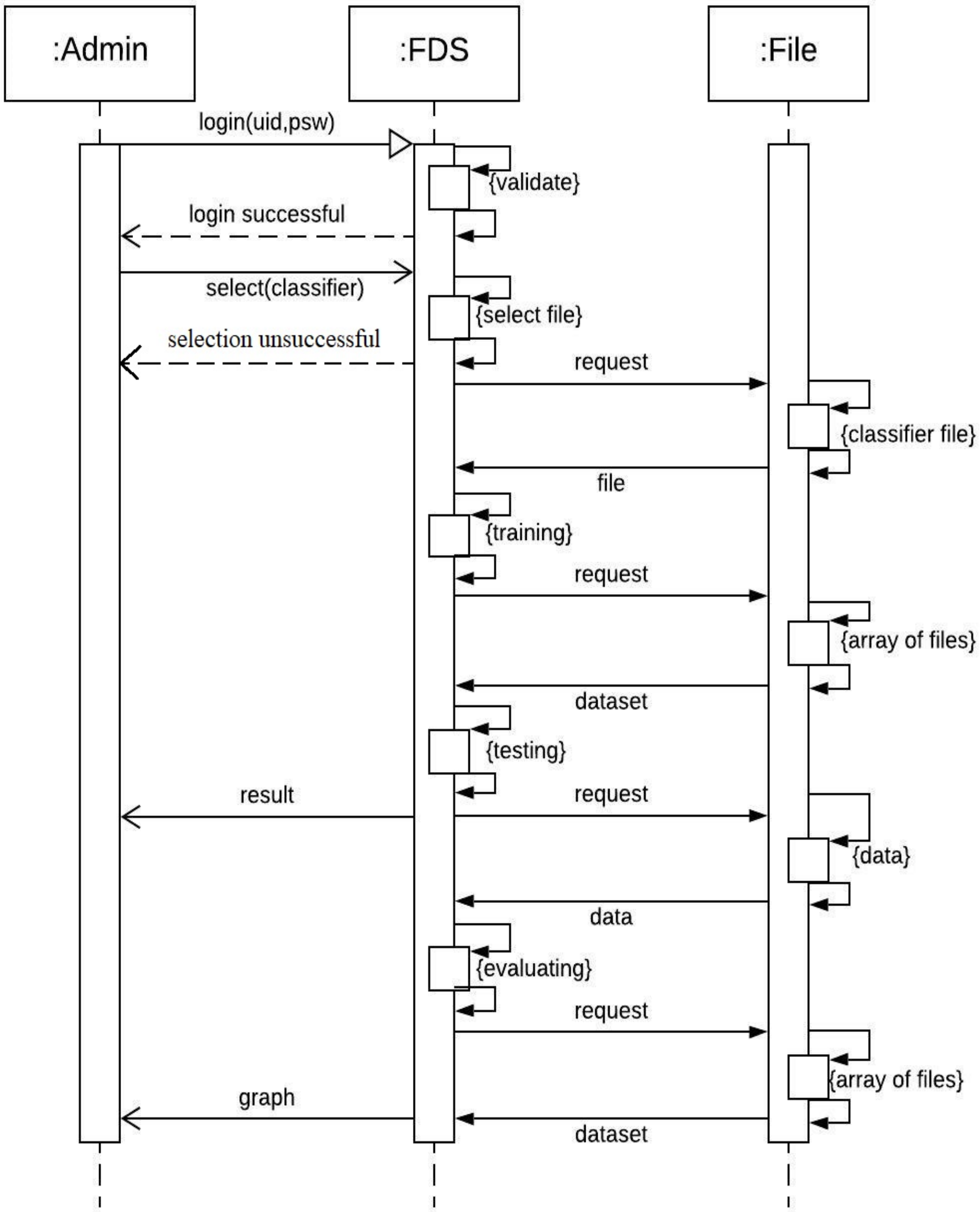

Figure 6.7: Sequence Diagram of Fraud Detection System

6.4.3 Activity Diagram

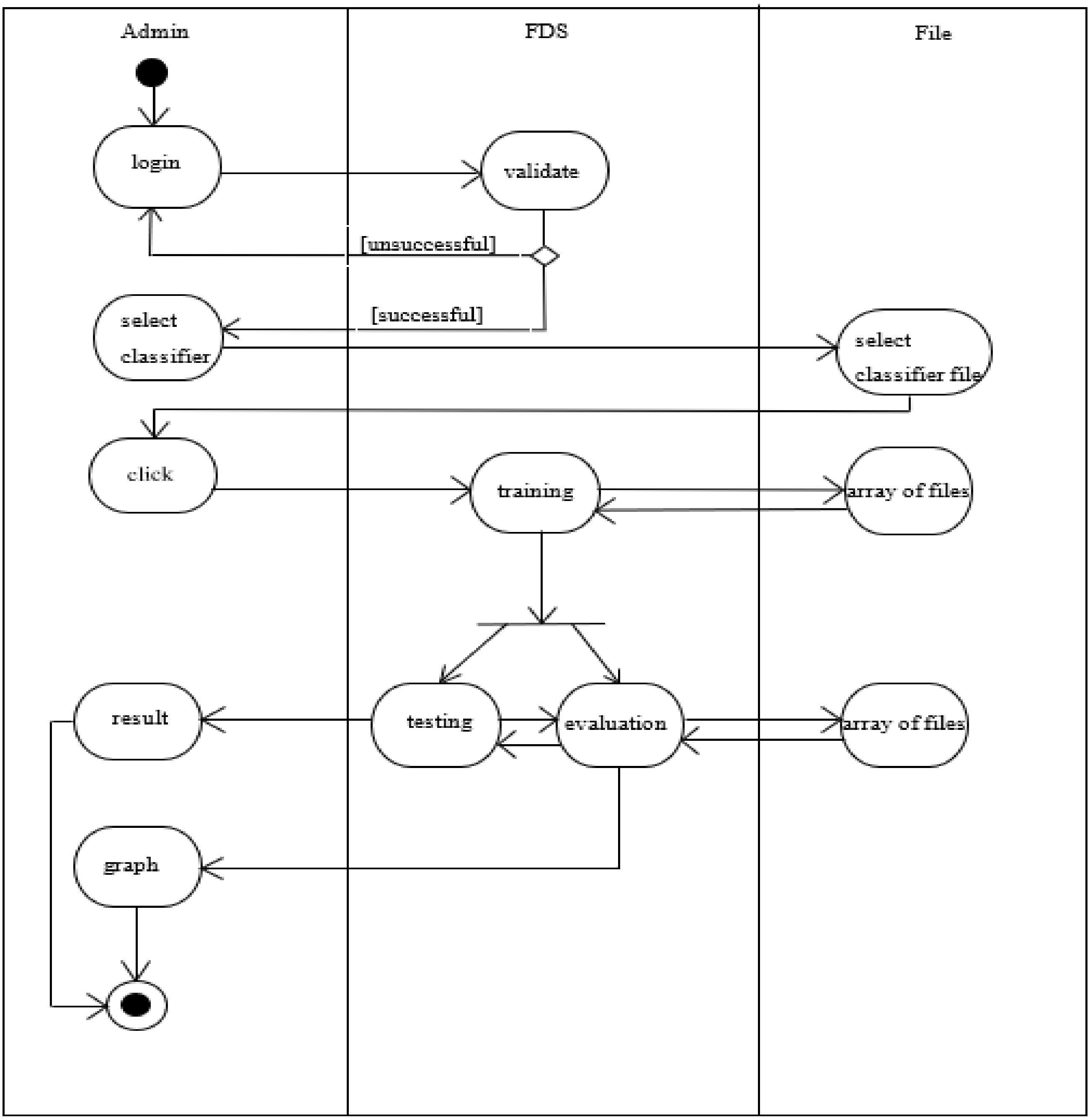

Figure 6.8: Activity Diagram of Fraud Detection System

Activity diagram portrays the arrangement of exercises that are conveyed by various items alongside control stream starting with one movement then to the next. Here the administrator signs in to the framework and on effective sign in the administrator can picked either to begin the testing or begin the preparation and assessment and on grouping the framework advises through the message whether the exchange is misrepresentation or not.

6.5 Modular Diagram

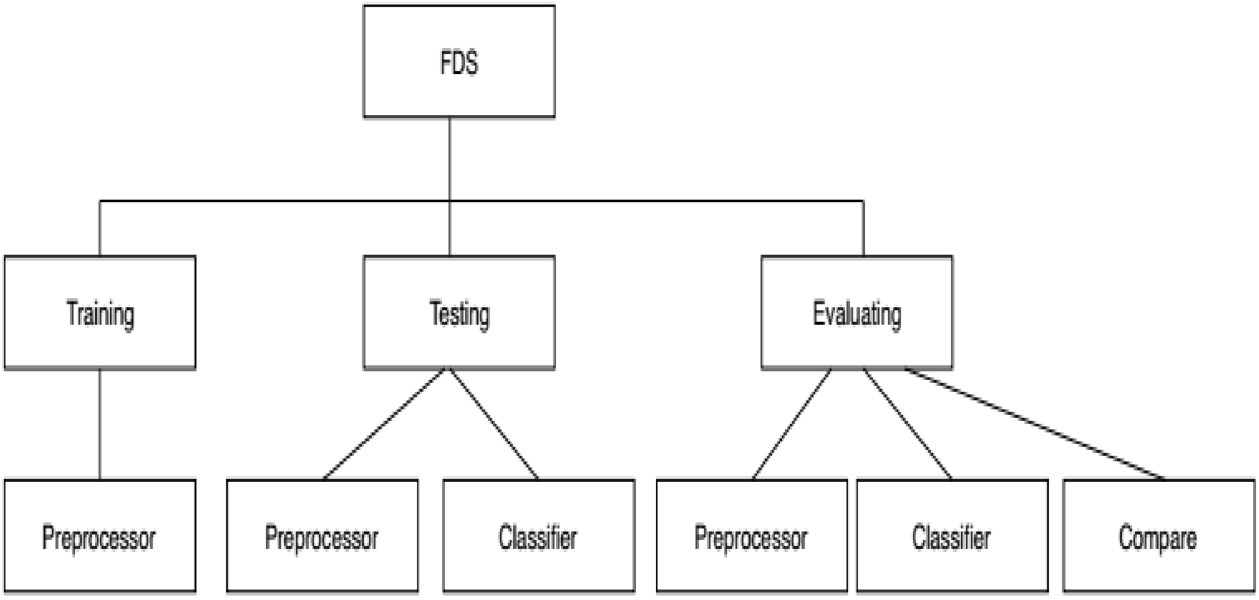

Figure 6.9: Modular Diagram of Fraud Detection System

Modular Diagram portrays how a framework is subdivided into littler modules. These modules can be made freely and after that can be utilized in various systems.The whole framework can be partitioned into modules like training,testing and evaluating.The preparing module incorporates pre-handling, testing module incorporates pre-handling and characterization and assessment incorporates pre-handling, order and comparision.

6.6 Data Flow Diagram

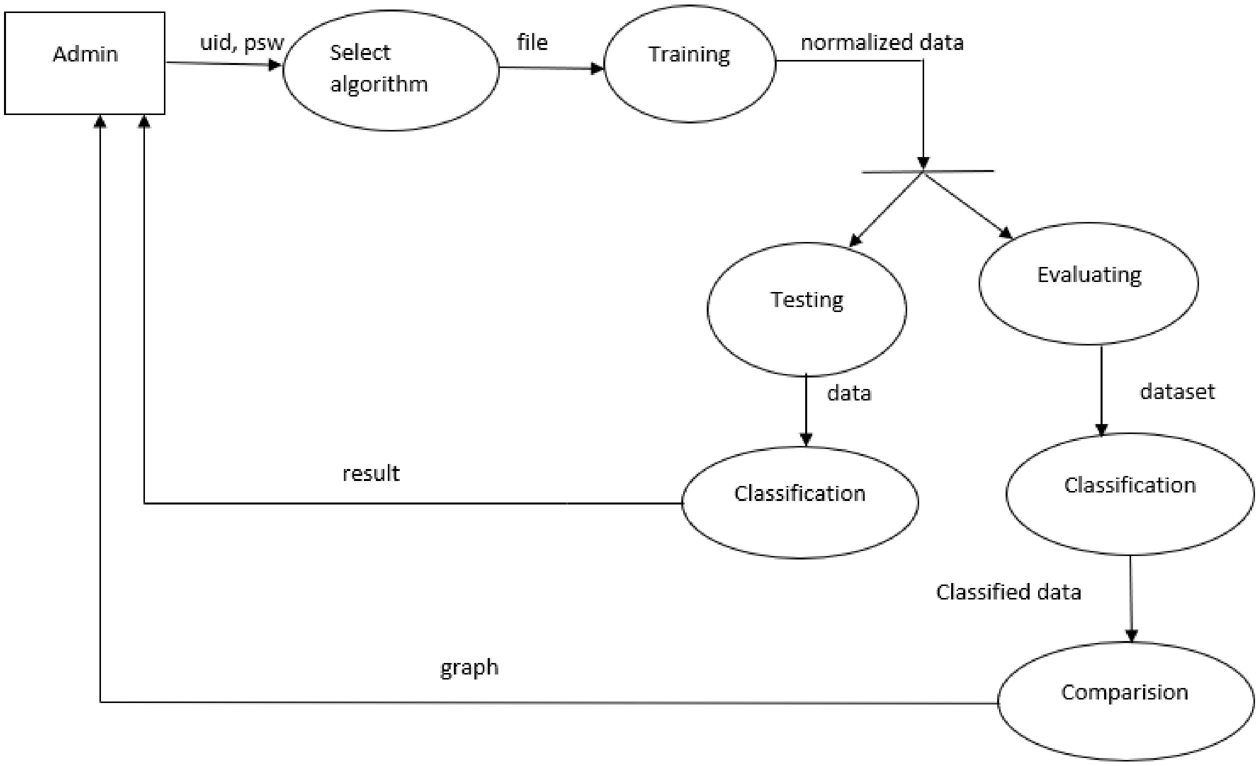

Figure 6.10: Data Flow Diagram of Fraud Detection System

The information stream display is graphical portrayal that demonstrates the progression of information between items or procedures. The square shape box documents as the article and the circle speaks to the capacity or procedure.

Chapter 7

Implementation

The most important and final phase in software development that has to be executed is implementation. Implementation involves realizing technical requirements or algorithms and converting it into a system, program or a software component with the help of computer programming and deploying it into the environment where it was desired to work.

The plan for implementation is made before the system has actually been implemented. The chapter show cases the conversion of each model plan into execution. The section below involves the steps for training and testing phases.

7.1 Algorithm for Login

Step 1: Start

Step 2: Enter the admin name and password

Step 3: if(username and password is valid)

Login

Else

Display error message

Step 4: Stop

7.2 Algorithm for Training

Step 1: Select the Algorithm

Step 2: Preprocessing the dataset

Step 3: Feed the dataset to the algorithm

Step 4: The features are saved for creating training model

Step 5: Check whether training is completed. If not, repeat Step 2 until training model is created

Step 6: Stop the Training when the task is completed

7.3 Algorithm for Testing

Step 1: Preprocessing the data

Step 2: Feed the data to the algorithm

Step 3: The system checks if the data is similar to the training model

Step 4: The result of the data will be displayed on the screen

Step 5: Stop the Testing when the task is completed

7.4 Algorithm for Evaluating

Step 1: Preprocessing the data

Step 2: Feed the dataset to the algorithm

Step 3: The system checks if each data is similar to the training model

Step 4: Output will be generated from the training model for each data

Step 5: Generated output is compared with the actual output

Step 6: The accuracy and confusion matrix will be obtained from the Step 5

Step 7: Stop the Evaluating when the task is completed

Chapter 8

Testing

Testing is a process where the system undergoes checks on the behaviour to satisfy the defined requirements. The process testing is performed by executing test cases and checking whether the behaviour of the system is positive or negative.

8.1 Types of Testing

The testing that are covered here are Unit testing and Integration testing.

8.1.1 Unit Testing

It is a procedure of testing every module exclusively to check the module meets its normal prerequisite making. Since the scope of this kind of testing is narrow, small pieces of codes are normally used. These kind of testing are the first type of testing usually executed by the developers. This testing is a like white box testing in light of the fact that the execution of the module is checked.

8.1.2 Integration Testing

When different modules of the framework has to be tested together, Integration testing comes in handy. It is a type of testing where different modules are put together and tested so as to meet the requirements of the system as whole. It requires more exertion since entire module has to be put together. They are conducted after unit testing and they normally starts with the interface specification. When they are conducted, they discover issues which rises when different modules interact with each other. It is a form of Black Box testing conducted along with White Box testing.

Test Case ID	Test Condition	Expected Outcome	Actual Outcome	Status
TC1	When user id and password are valid	Login Successful	Login Successful	PASS
TC2	When user id and password are invalid	Login Unsuccessful	Login Unsuccessful	PASS
TC3	When user does not enter id or password	Error message	Error message	PASS
TC4	User select the algorithm and train option	Model Trained	Model Trained	PASS
TC5	User select the test option for data-set	Status of test data-set	Status of test data-set	PASS
TC6	User select evaluate option for data-set	Gives confusion matrix and accuracy for testing data-set	Gives confusion matrix and accuracy for testing data-set	PASS

Table 8.1: Test Cases on Credit Card Fraud Detection System

Chapter 9

Results and Analysis

The experimental results of credit card fraudulent detection utilizing different learning models are shown in the confusion matrix.
The confusion matrix it's the method or the technique for plotting the introduction of a request computation.

	Class 1 Predicted	Class 2 Predicted
Class 1 Actual	TP	FN
Class 2 Actual	FP	TN

Figure 9.1: Confusion Matrix Structure

Here,

- Class 1 : Positive
- Class 2 : Negative

Definition of the Terms:

- Positive (P) : Observation is positive.
- Negative (N) : Observation is not positive.

- True Positive (TP) : Perception is certain, and is anticipated to be sure.
- False Negative (FN) : Perception is certain, and is anticipated to be not sure.
- True Negative (TN) : Perception is not certain, and is anticipated to be not sure.
- False Positive (FP) : Perception is not certain, and is anticipated to be sure.

Characterization exactness alone can be misdirecting in the event that you have an unequal number of perceptions in each class or on the off chance that you have multiple classes in your dataset.

Ascertaining a disarray network can give you a superior thought of what your order model is getting right and what sorts of blunders it is making. Classification accuracy is a proportion of right expectations to add up to forecasts made.

classification accuracy = correct predictions / total predictions

It is often presented as a percentage by multiplying the result by 100.

classification accuracy = correct predictions / total predictions * 100

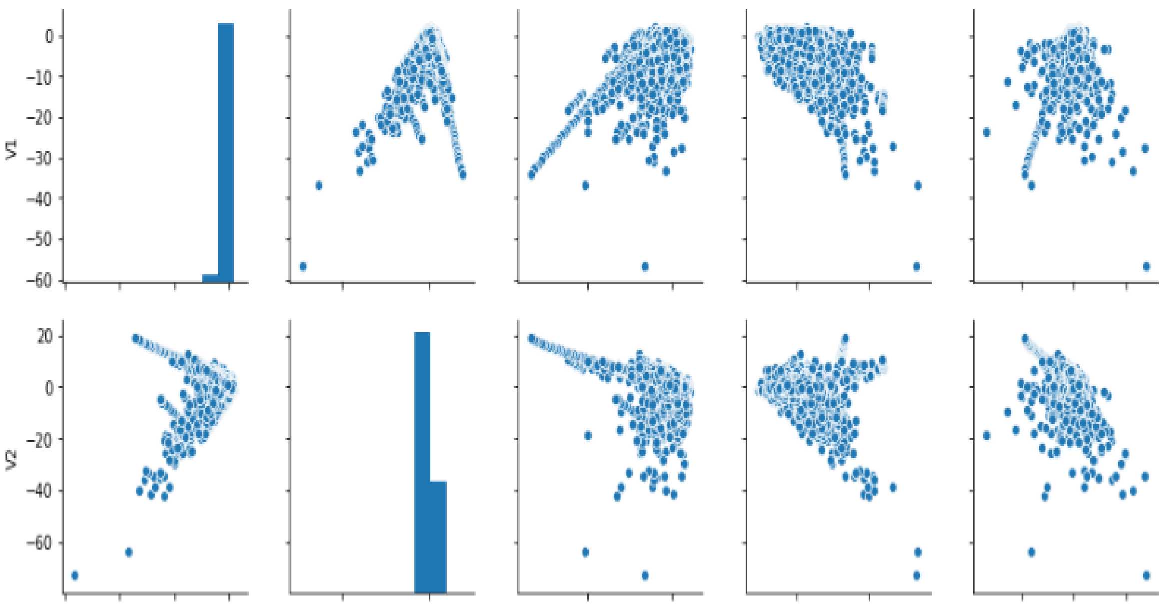

Figure 9.2: Bar-plot on dataset

A bar plot represents an estimate of central tendency for a numeric variable with the height of each rectangle and provides some indication of the uncertainty around that estimate using error bars. Bar plots incorporate 0 in the quantitative pivot range, and they are a decent decision when 0 is a significant incentive for the quantitative variable, and you need to make examinations against it.

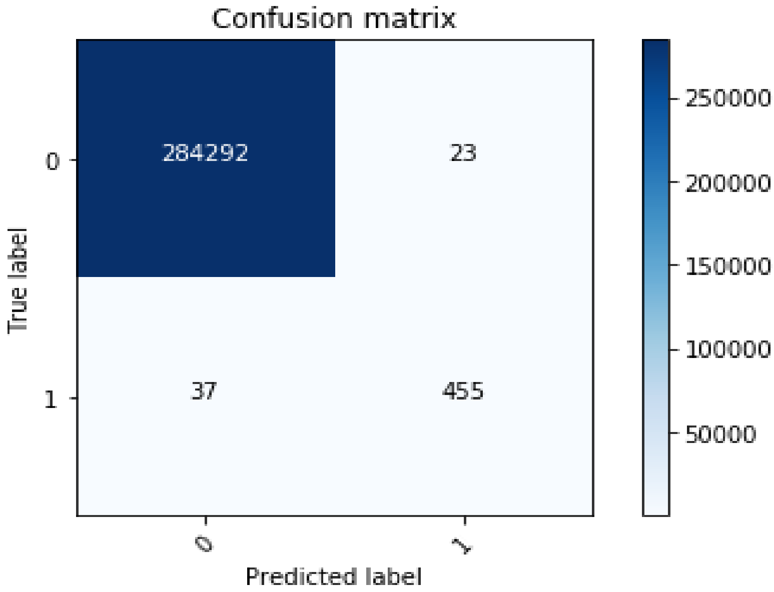

Figure 9.3: Confusion Matrix for Decision Tree

From the figure 9.3 the obtained accuracy is 99.929 and the confusion matrix value for true positive is 284294, the esteem for genuine negative is 23, the value for false positive is 37, and the value for false negative is 455.

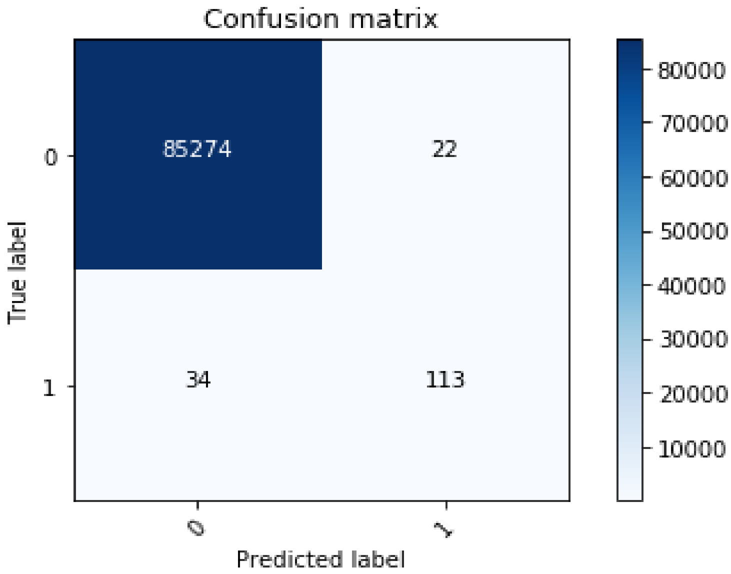

Figure 9.4: Confusion Matrix for Deep Learning

From the figure 9.4 the obtained accuracy is 99.939 and the confusion matrix value for true positive is 85274, the value for true negative is 22, the value for false positive is 34, and the value for false negative is 113.

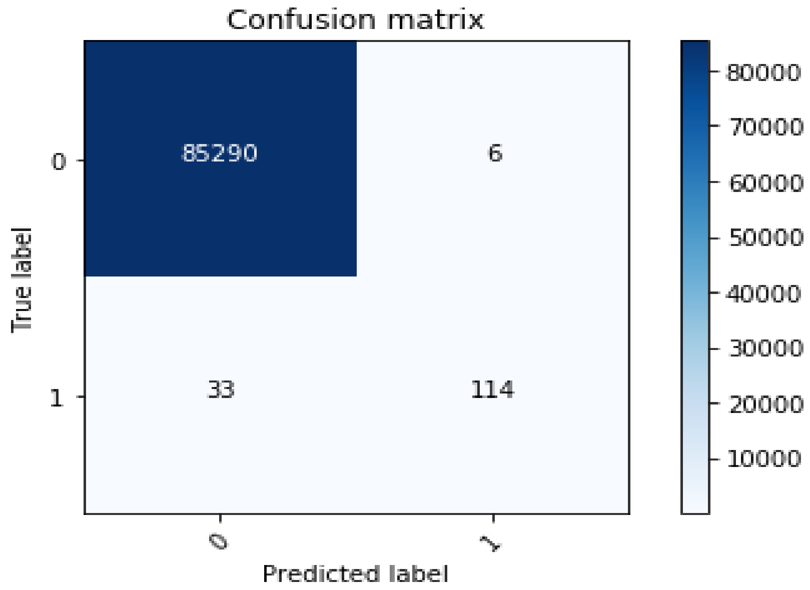

Figure 9.5: Confusion Matrix for Random Forest

From the figure 9.5 the obtained accuracy is 99.954 and the confusion matrix value for true positive is 85290, the esteem for genuine negative is 6, the value for false positive is 33, and the value for false negative is 114.

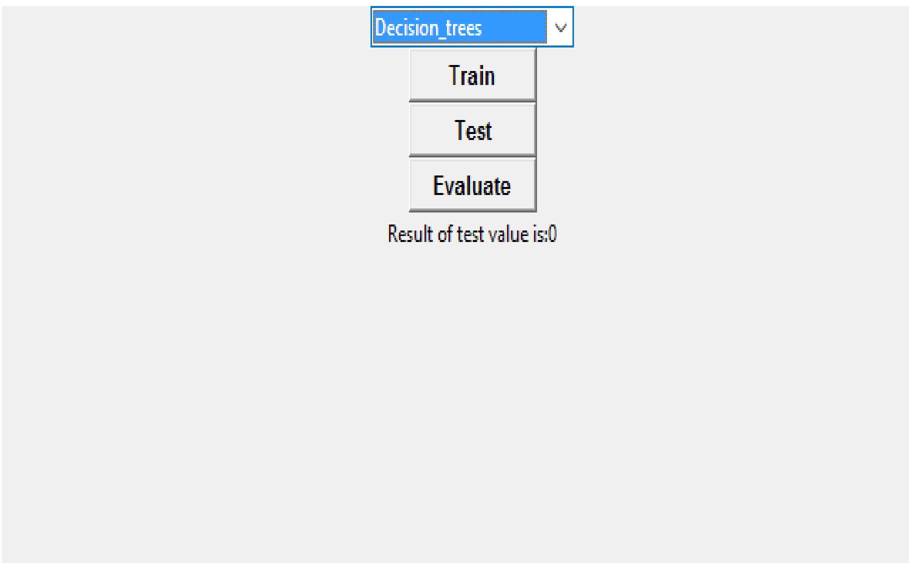

Figure 9.6: Decision tree result when test value is 0

We are giving the input as real time dataset with all the features to test the dataset. When the testing is done, output dataset matches with the real time features of the dataset. This transaction will be the genuine transaction and the result of the test case will show the value as 0.

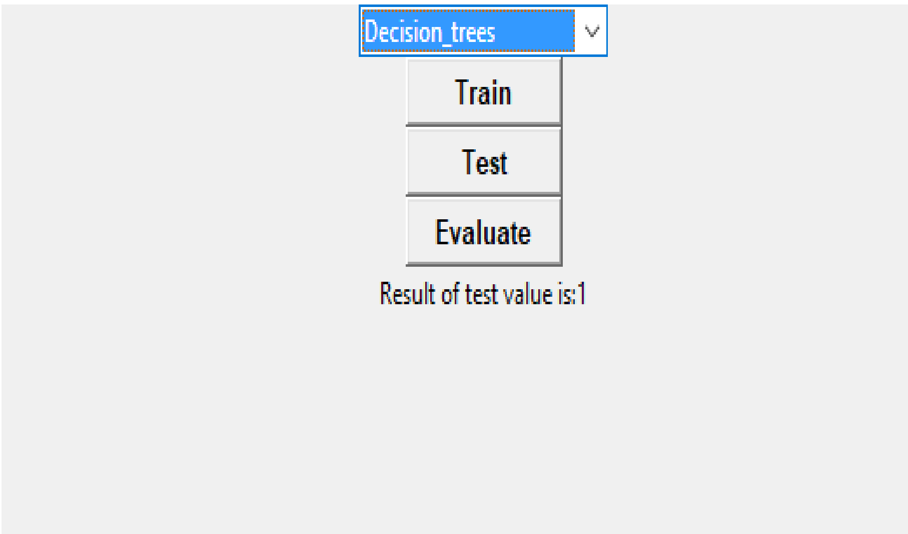

Figure 9.7: Decision tree result when test value is 1

We are giving the input as real time dataset with all the features to test the dataset. When the testing is done, output dataset does not matches with the real time features of the dataset. This transaction will be the fake transaction and the result of the test case will show the value as 1.

Chapter 10

Conclusion

The proposed project predicts whether the transaction is fraudulent transaction or genuine transaction based on the different features. The analyses of choosing the useful attributes and normalizing those attributes that will increase the performance of the model. By using this project we can predict the fraudulent transaction. Depending on the diagonal value of confusion matrix of all three learning models, decision tree has got highest value when compared with other. The accuracy of the random forest is better than the deep learning and decision tree.

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